

# Limits & Continuity

Rough def: limit of  $f(x)$  is  $L$  as  $x$  tends to  $c$  if  $f(x)$  tends to  $L$  as  $x$  tends to  $c$   
( $\lim_{x \rightarrow c} f(x) = L$ )  
e.g.  $\lim_{x \rightarrow 0} (2x+1) = 1$

1. Motivation/Background
2. Physical/geometrical intuition
3. Derivation/Proofs of theorem/formula
4. Applications

## Motivation 1: Instantaneous velocity of a moving object



$$y = s(t) = t^2 \quad (\text{Unit: m})$$

↑ Position of object at time  $t$

Instantaneous velocity at  $t=1$  ( $h$  is small)

Idea: find avg. velocity on time interval from 1 to  $1+h$

$$[1, 1+h] \quad h > 0$$

$$[1+h, 1] \quad h < 0$$

Change in  $y$

$$\frac{s(1+h) - s(1)}{h} = \frac{(1+h)^2 - 1^2}{h}$$

Change in  $x$

$$V(1) = \lim_{h \rightarrow 0} \frac{s(1+h) - s(1)}{h} = \lim_{h \rightarrow 0} \frac{(1+h)^2 - 1^2}{h}$$

Numerical:

| $h$  | avg. velocity on $[1, 1+h] = \frac{\Delta y}{\Delta x}$ |
|------|---------------------------------------------------------|
| 1    | 3                                                       |
| 0.1  | 2.1                                                     |
| 0.01 | 2.001                                                   |
| -0.1 | 1.9                                                     |

Algebraic way:

$$\lim_{h \rightarrow 0} \frac{(1+h)^2 - 1^2}{h} = \lim_{h \rightarrow 0} \frac{(1+h-1)(1+h+1)}{h} = \lim_{h \rightarrow 0} \frac{h(h+2)}{h} = \lim_{h \rightarrow 0} h+2 = 2$$

In general:  $y = S(t)$  is position func of moving object  
 Then instantaneous velocity at  $t = t_0$  is  $v(t_0) = \lim_{h \rightarrow 0} \frac{\Delta y}{\Delta t} = \lim_{h \rightarrow 0} \frac{S(t_0+h) - S(t_0)}{h}$

Motivation 2: Slope of tangent line of curve

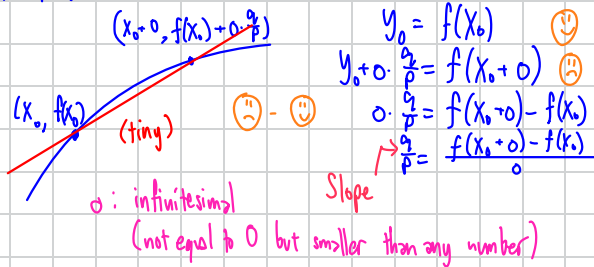


Why care about T-lines  $\rightarrow$  it is the best line to approximate the curve near point of tangency

Linearization

Slope of tangent line =  $\frac{\text{vertical velocity } q}{\text{horizontal velocity } p}$

Curve: trajectory of moving object (Newton)  
 $x = \text{time}$



$$\begin{aligned}
 f(x) &= x^2, x_0 = 1 \\
 y_0 &= x_0^2 = 1^2 = 1 \\
 y_0 + 0 \cdot \frac{q}{p} &= (x_0 + 0)^2 \\
 0 \cdot \frac{q}{p} &= (1 + 0)^2 - 1^2 = 0^2 + 2 \cdot 0 + 1 - 1 = 0^2 + 2 \cdot 0 \\
 \frac{q}{p} &= 0 + 2 = 2
 \end{aligned}$$

Leibniz: T-line is the line through a pair of infinitely close points on the curve  
 (Modern way  $\rightarrow$  T-line =  $\lim_{P \rightarrow Q}$  (secant line through P & Q) (not identical))

Slope of T-line at  $(x_0, f(x_0)) = L$



$$\lim_{h \rightarrow 0} \left( \frac{f(x_0+h) - f(x_0)}{h} \right) \frac{\Delta y}{\Delta x}$$

$$= \lim_{h \rightarrow 0} \frac{\Delta y}{\Delta x}$$

truncated neighborhood of  $x_0$



$$\frac{(1+h)^2 - 1^2}{h}$$

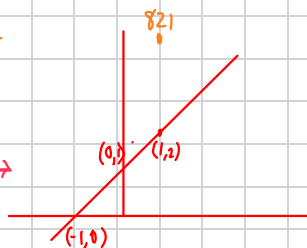
### Formal definitions of limit

Suppose  $f$  is defined in open interval about  $x_0$ , except possibly at  $x_0$  itself. If  $f(x)$  is arbitrarily close to the number  $L$ , as close as we wish, as long as  $x$  is close to, but not equal to  $x_0$ , enough, then we say the limit of  $f(x)$  is  $L$  as  $x$  approaches  $x_0$ .

e.g. 1  $f(x) = \frac{x^2 - 1^2}{x - 1} = x + 1$

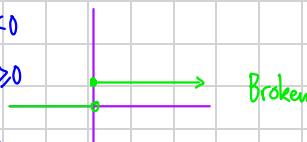
$$\lim_{x \rightarrow 1} f(x) = 2$$

e.g. 2  $g(x) = \begin{cases} \frac{x^2 - 1^2}{x - 1} & ; x \neq 1 \\ 821 & ; x = 1 \end{cases}$

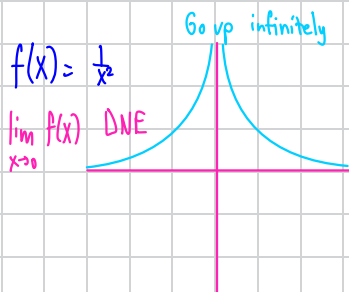


$\lim_{x \rightarrow 0} f(x)$  does not always exist

e.g.  $f(x) = \begin{cases} 0 & ; x < 0 \\ 1 & ; x \geq 0 \end{cases}$



$\lim_{x \rightarrow 0} f(x)$  does not exist



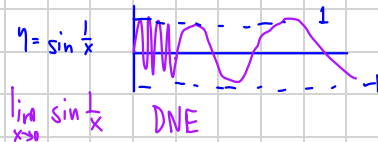
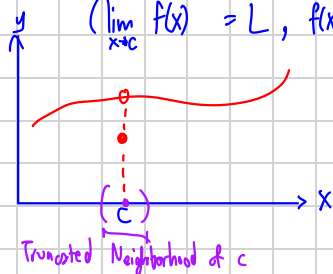
$y = \sin\left(\frac{1}{x}\right)$

$\lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right)$  DNE

(Dance too much)



Def. We say the limit of  $f(x)$  is the number  $L$  as  $x \rightarrow c$  if  $f(x)$  is arbitrarily close to  $L$  whenever  $x$  is close to (but not equal to)  $c$  enough. Write  $f(x) \rightarrow L$  when  $x \rightarrow c$  ( $\lim_{x \rightarrow c} f(x) = L$ ,  $f(x)$  is only required to be defined in a truncated neighborhood of  $c$ )

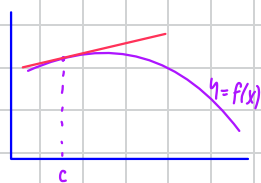


Important limits:  $y = s(t) \rightarrow$  position function of object moving on  $y$ -axis  

$$v(t_0) = \lim_{h \rightarrow 0} \frac{s(t_0+h) - s(t_0)}{h} \rightarrow \text{Instantaneous velocity at } t=t_0 = \frac{\Delta y}{\Delta t}$$

$$= \lim_{t \rightarrow t_0} \frac{s(t) - s(t_0)}{t - t_0}$$

Slope of tangent line to graph of  $y = f(x)$  at  $x = c$ :



$$\text{slope} = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \tan \theta$$

$$= \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$$

$\hookrightarrow$  rate of change of  $f$  at  $x=c$

2.1 (9) Slope/Tangent line's equation

$$y = x^2 - 2x - 3 \text{ at } P(2, -3)$$

Slope of curve at  $P$ :  $\lim_{h \rightarrow 0} \frac{(2+h)^2 - 2(2+h) - 3 - (-3)}{h} = \lim_{h \rightarrow 0} \frac{h^2 + 4h + 4 - 4 - 2h - 4 + 3 + 3}{h} = \lim_{h \rightarrow 0} \frac{h^2 + 2h}{h} = h + 2 = 2$

Equation of T-Line:  $y - (-3) = 2(x - 2)$   $y - b = m(x - a)$   
 $y = 2x - 7$

Limit laws: Suppose  $\lim_{x \rightarrow c} f(x)$ ,  $\lim_{x \rightarrow c} g(x)$  exist, then

$$\lim_{x \rightarrow c} [f(x) + g(x)] = \lim_{x \rightarrow c} f(x) + \lim_{x \rightarrow c} g(x) \quad \text{Sum rule}$$

$$\lim_{x \rightarrow c} (kf(x)) = k \lim_{x \rightarrow c} f(x) \quad k \text{ const}$$

$$\lim_{x \rightarrow c} f(x)g(x) = \lim_{x \rightarrow c} f(x) \cdot \lim_{x \rightarrow c} g(x)$$

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow c} f(x)}{\lim_{x \rightarrow c} g(x)} ; \lim_{x \rightarrow c} g(x) \neq 0$$

Quotient rule

$$\lim_{x \rightarrow c} (f(x))^n = \left( \lim_{x \rightarrow c} f(x) \right)^n$$

Power rule

$$\lim_{x \rightarrow c} (f(x))^{\frac{1}{n}} = \left( \lim_{x \rightarrow c} f(x) \right)^{\frac{1}{n}}$$

Root rule

$$\lim_{x \rightarrow c} x = c$$

const

$$\lim_{x \rightarrow c} x^2 = c^2$$

$$\lim_{x \rightarrow c} k = k$$

$$\lim_{x \rightarrow c} (a_2 x^2 + a_1 x + a_0) = a_2 c^2 + a_1 c + a_0$$

General case:  $P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$

$$\lim_{x \rightarrow c} P(x) = P(c)$$

Let  $P(x), Q(x)$  be polynomials  $\rightarrow \lim_{x \rightarrow c} \frac{P(x)}{Q(x)} = \frac{P(c)}{Q(c)} ; Q(c) \neq 0$

$$2.2 (15) \quad \lim_{x \rightarrow 2} \frac{2x+5}{11-x^3} = \frac{2 \cdot 2 + 5}{11 - 2^3} = \frac{9}{3} = 3$$

$$(20) \quad \lim_{z \rightarrow 4} \sqrt{z^2 - 10} = \sqrt{4^2 - 10} = \sqrt{6}$$

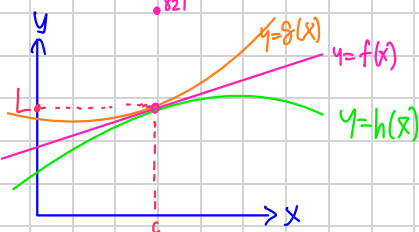
$$(42) \quad \lim_{x \rightarrow 4} \frac{4-x}{5-\sqrt{x^2+9}} = \lim_{x \rightarrow 4} \frac{4-x}{5-\sqrt{x^2+9}} \cdot \frac{5+\sqrt{x^2+9}}{5+\sqrt{x^2+9}} = \lim_{x \rightarrow 4} \frac{(4-x)(5+\sqrt{x^2+9})}{16-x^2}$$

$$= \lim_{x \rightarrow 4} \frac{5+\sqrt{x^2+9}}{4+x} = \frac{5+\sqrt{4^2+9}}{4+4} = \frac{5}{4}$$

Sandwich Theorem

$\rightarrow$  Suppose in a truncated neighborhood of  $c$ , we have  $h(x) \leq f(x) \leq g(x)$

Assume  $\lim_{x \rightarrow c} h(x)$  and  $\lim_{x \rightarrow c} g(x)$  exist and equal to each other



$\rightarrow \lim_{x \rightarrow c} f(x)$  exists at  $X=c$

Ex.  $\lim_{x \rightarrow 0} x \sin \frac{1}{x}$   $-1 \leq \sin \theta \leq 1$

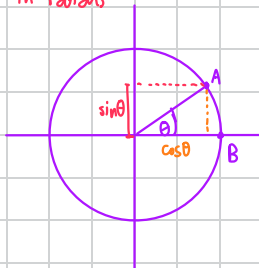
$$|x \sin \frac{1}{x}| = |x| |\sin \frac{1}{x}| \leq |x|$$

$$\therefore -|x| \leq x \sin \frac{1}{x} \leq |x|$$

$$\underbrace{h(x) \rightarrow 0 \quad f(x) \rightarrow 0 \quad g(x) \rightarrow 0}_{\theta \text{ in radians}} \quad \therefore \lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0$$

$$\lim_{\theta \rightarrow 0} \sin \theta = ?$$

$$\lim_{\theta \rightarrow 0} \cos \theta = ?$$



$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta + (1 - \cos^2 \theta) = \overline{AB}^2 < \overline{OB}^2 = \theta^2$$

$$\sin^2 \theta \leq \theta^2 \rightarrow |\sin \theta| \leq |\theta|$$

$$-|\theta| \leq \sin \theta \leq |\theta|$$

$$\nearrow = \sin 0$$

$$\lim_{\theta \rightarrow 0} : 0 \leq \lim_{\theta \rightarrow 0} \sin \theta \leq 0 \rightarrow \lim_{\theta \rightarrow 0} \sin \theta = 0$$

$$(1 - \cos^2 \theta) \leq \theta^2 \rightarrow |1 - \cos \theta| \leq |\theta|$$

$$-|\theta| \leq 1 - \cos \theta \leq |\theta|$$

$$\lim_{\theta \rightarrow 0} 0 \leq \lim_{\theta \rightarrow 0} 1 - \cos \theta \leq 0 \rightarrow \lim_{\theta \rightarrow 0} 1 - \cos \theta = 0$$

$$\lim_{\theta \rightarrow 0} \cos \theta = 1$$

$$\nearrow \cos 0$$

$$\lim_{\theta \rightarrow \theta_0} \sin \theta = ?$$

$$= \lim_{\theta \rightarrow \theta_0} \sin(\theta_0 + \theta - \theta_0)$$

$$= \lim_{h \rightarrow 0} \sin(\theta_0 + h) = \lim_{h \rightarrow 0} \sin h \cos \theta_0 + \sin \theta_0 \cos h$$

$$= \sin 0 \cos \theta_0 + \sin \theta_0 \cos 0$$

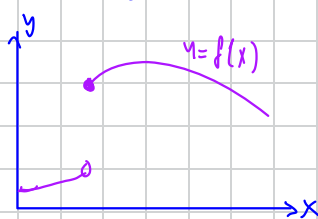
$$= \sin \theta_0$$

Similarly,  $\lim_{\theta \rightarrow \theta_0} \cos \theta = \cos \theta_0$

$$\lim_{\theta \rightarrow \theta_0} \tan \theta = \tan \theta_0$$

If  $\tan \theta_0$  defined

## 2.4 One-sided limits



$$\text{Right-limit: } \lim_{x \rightarrow c^+} f(x)$$

$$\text{Left-limit: } \lim_{x \rightarrow c^-} f(x)$$

$$\text{Theorem: } \lim_{x \rightarrow c} f(x) \text{ iff } \lim_{x \rightarrow c^+} f(x) = \lim_{x \rightarrow c^-} f(x) \text{ exist and equal}$$

$$\text{Important: } \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta}$$

$$\text{First } \lim_{\theta \rightarrow 0^+} \frac{\sin \theta}{\theta}$$



$$\text{Observe: Area of } \triangle OAB < \text{Area of sector OAB} < \text{Area of } \triangle OAC$$

$$\frac{1}{2} \cdot \sin \theta \cdot 1 < \frac{\theta}{2} < \frac{1}{2} \cdot 1 \cdot \tan \theta \rightarrow \frac{\sin \theta}{\cos \theta} < \frac{\theta}{\cos \theta} < \frac{1}{\sin \theta}$$

for small  $\theta$

$$\theta \cos \theta < \sin \theta < \theta$$

$$\cos \theta < \frac{\sin \theta}{\theta} < 1$$

$$\rightarrow 1 \quad \theta \rightarrow 1$$

(Squeeze theorem)

$$\therefore \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$$

$$\begin{aligned}\cos\theta &= \cos\left(\frac{\theta}{2} + \frac{\theta}{2}\right) = \cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2} = 1 - 2\sin^2\frac{\theta}{2} \\ \lim_{\theta \rightarrow 0} \frac{1 - \cos\theta}{\theta} &= \lim_{\theta \rightarrow 0} \frac{1 - 1 + 2\sin^2\frac{\theta}{2}}{\theta} = 2 \lim_{\theta \rightarrow 0} \frac{\sin^2\frac{\theta}{2}}{\theta} \\ &= 2 \lim_{\theta \rightarrow 0} \frac{\left(\frac{\sin^2\frac{\theta}{2}}{\left(\frac{\theta}{2}\right)^2}\right) \cdot \theta}{4} \\ &= 2 \cdot 1 \cdot 0 = 0\end{aligned}$$

$$\lim_{x \rightarrow 0} \frac{\tan x}{x} = \lim_{x \rightarrow 0} \frac{\sin x \cdot 1}{\cos x \cdot x} = 1$$

$$\lim_{x \rightarrow 0} \frac{\tan 3x}{x} = 3 \cdot \lim_{x \rightarrow 0} \frac{\tan 3x}{3x} = 3 \cdot 1 = 3$$

## 2.5 Continuity

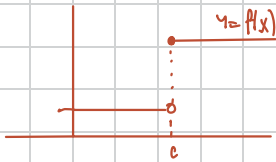
Definition We say  $f(x)$  is **continuous** at  $x=c$  if

- $\lim_{x \rightarrow c} f(x)$  exist ;
- $\lim_{x \rightarrow c} f(x) = f(c)$

If not, we say  $f(x)$  is **discontinuous** at  $x=c$

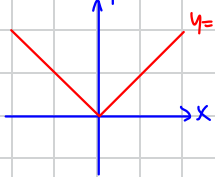
**right-continuous**:  $\lim_{x \rightarrow c^+} f(x)$  exist,  $\lim_{x \rightarrow c^+} f(x) = f(c)$

**left-continuous**:  $\lim_{x \rightarrow c^-} f(x)$  exist,  $\lim_{x \rightarrow c^-} f(x) = f(c)$



$\rightarrow f(x)$  is discontinuous at  $x=c$ , but is right-continuous at  $x=c$

E.g. 1  $f(x) = |x|$



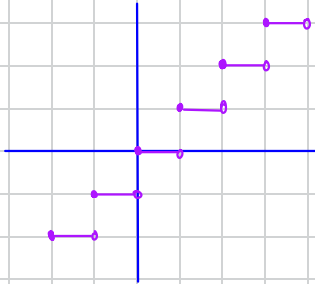
$y=|x|$  continuous everywhere

E.g. 2 Polynomials  $P(x) \rightarrow$  continuous everywhere

E.g. 3  $\frac{P(x)}{Q(x)} \rightarrow$  continuous at  $c$  if  $Q(c) \neq 0$

E.g. 4  $\sin X, \cos X \rightarrow$  continuous at every point in their domain  
 $\tan X, \cot X$   
 $\sec X, \csc X$

E.g. 5  $f(x) = \text{largest integer } \leq x$   
 $[0.5] = 0$   $[1.5] = 1$   $[-0.001] = -1$



Discontinuous at all integers  
Right-continuous at all integers  
Not left-continuous

HW Next wk 2.1, 2.2, 2.4, 2.5 ~~2.3~~ ~~2.6~~  
QUIZ